

Mucoease Plus

 Syrup

DESCRIPTION

Clear, yellow syrup.

COMPOSITION

Each 5 mL contains Carbocisteine 100 mg and Promethazine HCl 2.5 mg.

PHARMACODYNAMICS

Mucoease Plus Syrup combines the mucolytic and muco-regulatory properties of carbocisteine with the antihistaminic and antitussive properties of promethazine.

Carbocisteine is a mucolytic agent that reduces goblet cell hyperplasia in the management of respiratory disorders associated with abnormally excessive or viscous mucus secretion.

Promethazine is an H1 antihistamine, phenothiazine with an aliphatic side chain, characterized by:

- A marked sedative effect at standard doses as a result of histaminergic and adrenolytic activity.
- An anticholinergic effect causing peripheral adverse effects
- A peripheral adrenolytic effect, possibly affecting hemodynamics (risk of postural hypotension).

All antihistamines are opposed to the effects of histamine, particularly on the skin, bronchi, intestine and blood vessels, through more or less reversible competitive antagonism. Most have an antitussive effect, which is mild in itself but potentiates the effects of central morphine antitussive as well as those of other bronchodilators such as sympathomimetic amines, with which they are often combined.

PHARMACOKINETICS

Carbocisteine is rapidly and well absorbed after an oral dose and peak plasma concentrations occur about 2 hours. Bioavailability is low, less than 10% of the dose administered, probably because of intraluminal metabolism and a marked first-pass hepatic effect. Carbocisteine penetrates into lung tissues and respiratory mucus. Carbocisteine is metabolised via acetylation, decarboxylation and sulfoxidation by genetic polymorphism. It undergoes extensive

hepatic-first pass metabolism via hydroxylation by CYP2D6 and N-demethylation by CYB2B6. The elimination half-life is approximately two hours. Carbocisteine is excreted via urine, as unchanged drug and metabolites.

The bioavailability of promethazine is between 13% and 40%. 1.5-3 hours are necessary before peak plasma concentration is reached. The volume of distribution is high, because of the substance liposolubility, around 15 L/kg. The fraction bound to plasma proteins reaches 75 - 80%. The half-life is between 10 and 15 hours. Metabolism consists of sulfoxidation followed by demethylation. Sulfoxidation may be governed by genetic polymorphism. Renal clearance represents less than 1% of total clearance, and approximately 1% of the amount of promethazine administered is found in an unchanged form in the urine. The metabolites found in the urine, particularly sulfoxide, represent approximately 20% of the dose. It appears to penetrate into lung tissue and respiratory mucus.

Physiopathological variation: risk of accumulation of antihistamines in patients with kidney or liver insufficiency.

INDICATIONS

For the relief of cough due to cold and allergy.

CONTRAINDICATIONS

Mucoease Plus Syrup is contraindicated for:

- Children less than 2 years of age.
- Patients with active peptic ulceration or gastroduodenal ulcer.
- Hypersensitivity to active substances or any of the excipients.
- Hypersensitive or to have had an idiosyncratic reaction to promethazine or to other phenothiazines.
- Patients in comatose states or suffering from CNS depression of any cause.
- Antihistamines are contraindicated for use in the treatment for the symptoms of lower respiratory tract including asthma.
- Patients taking monoamine oxidase inhibitors up to 14 days previously.
- History of agranulocytosis.

PRECAUTIONS

Related to the presence of carbocisteine

- Caution is recommended for,
- Patients suffering from gastroduodenal ulcers, as mucolytics have the capacity to destroy the gastric mucosal barrier.
- Asthmatic patients with a history of bronchospasm.
- Patients with severe respiratory failure.
- Debilitated patients. By decreasing cough reflex there is a risk of obstruction of the airways as a consequence of the secretions amount increase.

Related to the presence of promethazine

- Use with care in closed-angle glaucoma (acute pain of the eye with visual disturbances) and in cases of prostate disorders, stenosing peptic ulcer, pyloroduodenal obstruction and bladder-neck obstruction.
- Use with care in cardiovascular disease or with severe liver and/or kidney insufficiency.
- The used of this medication should be avoided in patients with compromised respiratory functions (e.g. COPD and sleep apnea).

- Use with caution in persons with seizure disorders or in persons who are using concomitant medications, such as narcotics or local anesthetics, which may also affect seizure threshold.

Promethazine should be used with caution in patients with bone-marrow depression. Leukopenia and agranulocytosis have been reported usually when promethazine has been used in association with other known marrow-toxic agents.

A potentially fatal symptom complex sometimes referred to as Neuroleptic Malignant Syndrome (NMS) has been reported in association with promethazine alone or in combination with antipsychotic drugs. Clinical manifestations of NMS are hyperpyrexia, muscle rigidity, altered mental status and evidence of autonomic instability (irregular pulse or blood pressure, tachycardia, diaphoresis and cardiac dysrhythmias).

Promethazine may thicken or dry lung secretions and impair expectoration. It should therefore be used with caution in patients with asthma, bronchitis or bronchiectasis.

Promethazine may mask the warning signs of ototoxicity caused by ototoxic drugs e.g. salicylates. It may also delay the early diagnosis of intestinal obstruction or raised intracranial pressure through the suppression of vomiting.

- Photosensitizing.

It is irrational to combine an expectorant or a mucolytic with this antitussive medicine. Before prescribing an antitussive treatment, it is necessary to establish the causes of cough which require specific treatment. If the cough resists to the antitussive treatment administered at standard dosage, the dosage should not be increased but the clinical situation should be reassessed.

Use in children: If used in children, the possibility of bronchial asthma or gastroesophageal reflux should be ruled out before using promethazine as an antitussive. The use of promethazine should be avoided in children and adolescents with signs and symptoms of Reye's syndrome.

Use in elderly: Promethazine HCl should be used with care in elderly patients presenting with: Increased sensitivity to orthostatic hypotension, dizziness, sedation, chronic constipation (risk of paralytic ileus), possible prostate hypertrophy, certain cardiovascular disorders due to tachycardic and hypotensive effects of phenothiazines and cases of severe liver and/or kidney insufficiency. In general, dose selection for an elderly patient should be cautious, usually starting at the low end of the dosing range, reflecting the greater frequency of decreased hepatic, renal or cardiac function, and of concomitant disease or other drug therapy. Sedating drugs may cause confusion and over-sedation in the elderly. Thus, elderly patients generally should be started on low doses of promethazine and observed closely.

Related to the presence of excipients:

This medicinal product contains 3g of sucrose per 5ml, this must be taken account daily allowance in patients following a low-carbohydrates diet or in diabetic patients. This medicinal contains sodium. It contains 25mg of sodium per 5ml of syrup. This should be taken into account in patients on a strict sodium diet. This medicinal product contains methylparaben and cause allergic reactions (possibly delayed).

KEEP ALL MEDICINES OUT OF THE REACH OF CHILDREN

WARNING

This product contains Promethazine hydrochloride. It should not be used in paediatric patients less than 2 years of age because of the potential for fatal respiratory depression. For children 2 to 6 years of age to be used with caution and on doctor's or pharmacist's advice.

Productive coughs, with represent a fundamental element in the bronchial and pulmonary defences, should not be suppressed. It is irrational to combine an expectorant or a mucolytic with this antitussive medicine.

Before prescribing an antitussive, it is advisable to ensure that the underlying causes of the cough which require specific treatment, have been investigated.

If the cough does not respond to the antitussive treatment administered at the normal dosage, the dosage should not be increased but the clinical situation should be reassessed.

Mucolytic agents may induce severe bronchial congestion in infants. Their bronchial mucus drainage capacities are limited due to the physiological characteristics of their bronchial tree. Mucolytics should therefore not be used in infants. Treatment should be re-evaluated if symptoms or the disease persist or worsen.

This medicinal product contains sucrose. It is not recommended for use in patients with fructose intolerance, glucose or galactose malabsorption syndrome of sucrose-isomaltase deficiency.

PREGNANCY, LACTATION AND EFFECTS ON ABILITY TO DRIVE AND USE MACHINE.

PREGNANCY

Mucoease Plus Syrup should be avoided during the first trimester of pregnancy and should only be prescribed thereafter is necessary. It is also limited to occasional use in the third trimester.

If this medicine is used at the end of pregnancy, monitoring the nerves and digestive function in the new born appears to be warranted.

LACTATION

Since antihistamines are excreted into breast-milk, although in small quantities, and owing to the marked sedative properties of promethazine, this medicinal product should be avoided in breast-feeding women.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Patients who drive or use machines should be warned of the risk of drowsiness associated with this medicinal product, particularly at the start of treatment.

This is exacerbated by alcoholic beverages or medicinal products containing alcohol.

INTERACTION

Drug interactions include:

1. Alcohol - Alcohol enhances the sedative effect of antihistamines. Impaired alertness can make driving or operating machinery dangerous. The consumption of alcoholic beverages and alcohol-containing medicinal products is to be avoided.

2. Other CNS depressants - Promethazine may increase, prolong, or intensify the sedative action of other CNS depressants. Therefore, such agents should be avoided or administered in reduced dosage to patients receiving promethazine. When given concomitantly with promethazine, the dose of barbiturates should be reduced by at least one-half, and the dose of narcotics should be reduced by one-quarter to one-half. Dosage must be individualized. Excessive amounts of promethazine relative to a narcotic may lead to restlessness and motor hyperactivity in the patient with pain. These symptoms usually disappear with adequate control of the pain.

3. Sultopride - Increased risk of ventricular rhythm disorders, particularly torsades de pointes, by addition of electrophysiological effects.

4. Atropine and other atropine substances (Impramine antidepressants, anticholinergic antiparkinsonians, atropine antispasmodics, disopyramide, phenothiazine neuroleptics), addition of adverse atropine effects such as urinary retention, constipation, dry mouth.

5. Epinephrine - Epinephrine should not be used to treat hypotension associated with promethazine overdose because of the potential for promethazine to reverse epinephrine's vasopressor effect.

6. Anticholinergics - Concomitant use of other agents with anticholinergic properties should be undertaken with caution.

7. Monoamine Oxidase Inhibitors (MAOI) - Drug interactions, including an increased incidence of extrapyramidal effects, have been reported when some MAOI and phenothiazines are used concomitantly.

ADVERSE REACTIONS

Related to carbocisteine:

Respiratory Disorders:

Risk of severe bronchial congestion (particularly in infants and in some patient incapable of effective expectoration).

Skin and Subcutaneous Tissue Disorders:

Allergic skin reactions and severe cutaneous adverse reactions (SCAR) e.g. erythema multiforme, Steven-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN). In most of these cases reported at least one other drug was administered at the same time, which may have possibly enhanced the described mucocutaneous effects.

Cardiac Disorders:

Palpitation.

Endocrine Disorders:

Hypothyroidism.

Gastrointestinal Disorders:

Possibility of gastrointestinal intolerance symptoms such as gastric pain, diarrhoea, nausea, vomiting, stomach pain, gastrointestinal bleeding and gastric discomfort.

Immune System Disorders:

Anaphylactic/ anaphylactoid reaction.

Musculoskeletal and Connective Tissue Disorders:

Muscle pain.

Nervous System Disorders:

Headache, dizziness, urinary incontinence and palpitations

Related to promethazine:

Autonomic Nervous System Disorders:

Sedation or drowsiness (more marked at the beginning of treatment), somnolence, dryness of the mucosae constipation, accommodation difficulties (rarely), mydriasis, cardiac palpitations, risk of urinary retention, orthostatic hypotension, equilibrium disorders, vomiting, dizziness, impaired memory or concentration, motor incoordination, tremor (more frequent in elderly patients), mental confusion, hallucinations (more rarely, and particularly in infants), excitation effects (agitation, nervousness, insomnia), blurred vision, extrapyramidal symptoms (such as oculogyric crisis, torticollis, and tongue protrusion), lassitude, tinnitus, fatigue, euphoria, nervousness, diplopia, insomnia, convulsive seizures, catatonic-like states and hysteria.

Cardiovascular Disorders:

Increased or decreased blood pressure, tachycardia, bradycardia, faintness.

Skin and Subcutaneous Tissue Disorders:

Allergic skin reactions such as erythema, eczema, pruritus, purpura, urticaria, oedema (more rarely Quincke's oedema), dermatitis and photosensitivity.

Haematological Disorders:

Leukopenia, neutropenia, exceptional cases of agranulocytosis, thrombocytopenia, haemolytic anaemia and thrombocytopenic purpura.

Gastrointestinal Disorders:

Dry mouth (rarely), nausea, and jaundice.

Respiratory Disorders:

Asthma, nasal stuffiness, respiratory depression (potentially fatal) and apnoea (potentially fatal).

Paradoxical Reactions:

Hyperexcitability and abnormal movements have been reported in patients following a single administration of promethazine HCl. Consideration should be given to the discontinuation of promethazine HCl and to the use of other drugs if these reactions occur. Respiratory depression, nightmares, delirium, and agitated behaviour have also been reported in some of these patients.

Other:

Angioneurotic oedema and neuroleptic malignant syndrome (potentially fatal).

DOSAGE AND ADMINISTRATION

Usual dosage:

Adults:

3 to 4 tablespoonsfuls per day, in divided doses.

Children:

25 to 30 months: 3 to 4 of the 5ml measuring spoonfuls per day, in divided doses.

30 months to 12 years: 4 to 6 of the 5ml measuring spoonfuls per day, in divided doses.

12 to 15 years: 6 to 9 of the 5ml measuring spoonfuls per day, in divided doses.

IN ALL CASES YOU SHOULD COMPLY STRICTLY WITH YOUR DOCTORS'S PRESCRIPTION.

INCOMPATIBILITIES:

None known to date

OVERDOSAGE:

Symptoms and treatment of an overdose of carbocisteine:

Gastrointestinal disturbance is the most likely symptoms of carbocisteine overdose. Gastric lavage may be beneficial, followed by observation.

Symptoms and treatment of an overdose of promethazine:

Signs and symptoms of overdosage with promethazine range from mild depression of the central nervous system and cardiovascular system to profound hypotension, respiratory depression, unconsciousness, coma and sudden death. Other reported reactions include drowsiness, hypothermia, tachycardia, nausea, fever, stiffness of muscle, urinary retention, hyperreflexia, ataxia, athetosis, and extensor-plantar reflexes (Babinski reflex). Stimulation may be evident, especially in children and geriatric patients. Convulsions may rarely occur. A paradoxical-type reaction has been reported in children receiving single doses of 75 mg to 125 mg orally, characterized by hyperexcitability and nightmares or hallucination. Atropine like signs and symptoms – dry mouth, fixed, dilated pupils, flushing, as well as gastrointestinal symptoms may occur.

Treatment: Treatment of overdosage is essentially symptomatic and supportive. Only in cases of extreme overdosage or individual sensitivity do vital signs, including respiration, pulse, blood pressure, temperature and electrocardiogram (EKG/ECG), need to be monitored. Activated charcoal orally by lavage may be given, or sodium or magnesium sulphate orally as cathartic. Attention should be given to the re-establishment of adequate respiratory exchange through provision of patent airway and institution of assisted or controlled ventilation. Diazepam may be used to control convulsions. Acidosis and electrolyte losses should be corrected. Note that any depressant effects of promethazine are not reversed by naloxone. Avoid analeptics which may cause convulsions.

The treatment of choice for resulting hypotension is administration of intravenous fluids, accompanied by repositioning if indicated. In the event that vasopressors are considered for the management of severe hypotension which does not respond to intravenous fluid, the administration of norepinephrine

EPINEPHRINE SHOULD NOT BE USED, since its use in patients with partial adrenergic blockade may further lower the blood pressure. Extrapyramidal reactions may be treated with anticholinergic anti-parkinson agents, diphenhydramine, or barbiturates. Oxygen may also be administered. Limited experience with dialysis indicates that it is not helpful.

ROUTE OF ADMINISTRATION:

Oral.

STORAGE CONDITIONS

Store below 30°C.

Protect from light.

Keep container tightly closed

SHELF-LIFE

Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION

120 mL in amber glass bottle.

Date of revision: 05th August 2019

PRODUCT REGISTRATION HOLDER/

MANUFACTURER:

PHARMANIAGA MANUFACTURING

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